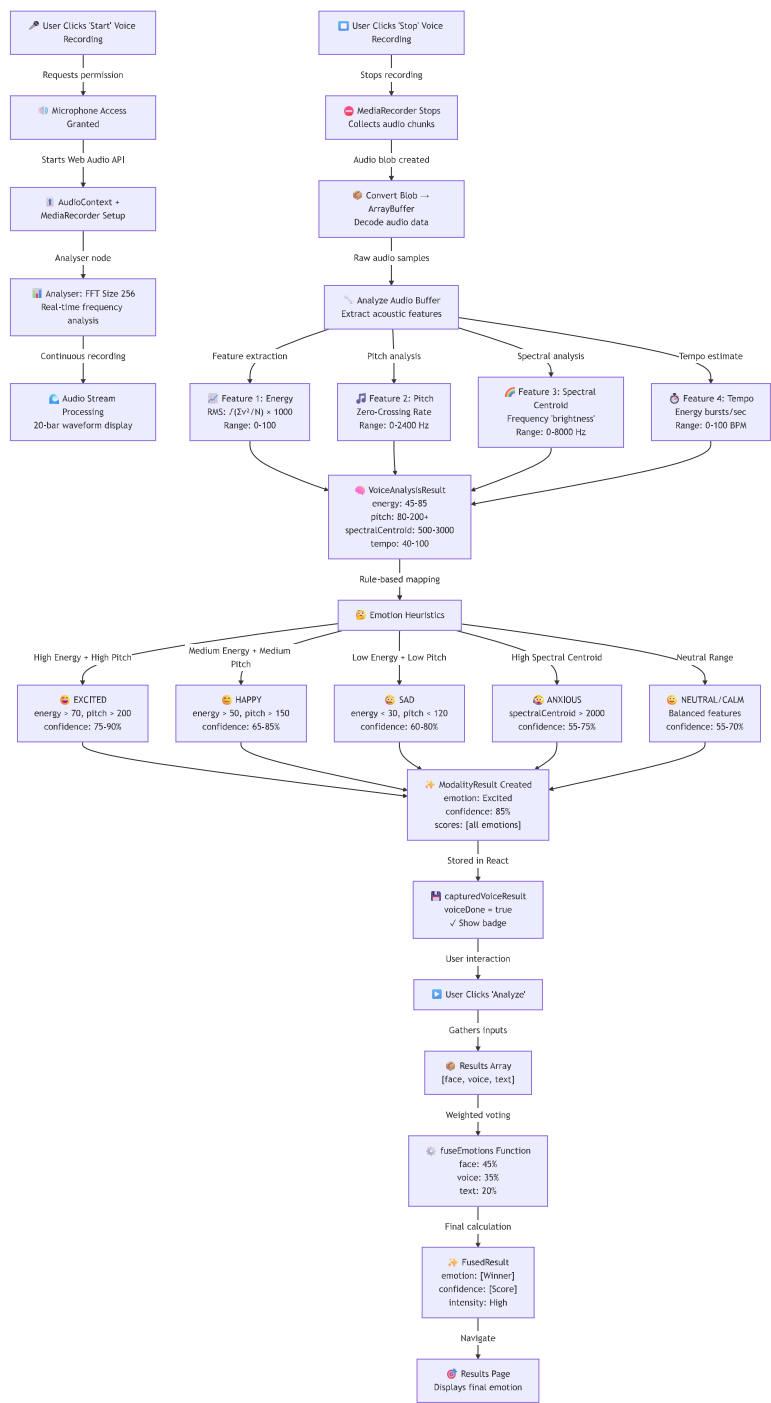






Audio Features Extracted

1. **Energy (RMS):** Overall volume/intensity of speech
2. **Pitch (ZCR):** How many times signal crosses zero per second
3. **Spectral Centroid:** "Brightness" of the sound
4. **Tempo:** Speaking rate (bursts per second)

```
energy =  $\sqrt{(\sum \text{samples}^2 / N) \times 1000}$  pitch  
= zero_crossings / length centroid =  
 $\sum(\text{frequency} \times \text{magnitude}) / \sum(\text{magnitude})$ 
```



Technology Stack

- **MediaRecorder API:** Captures audio stream
- **Web Audio API:** Decodes and analyzes audio
- **AudioContext:** Processes raw audio data
- **AnalyserNode:** Real-time frequency analysis
- **FFT (Fast Fourier Transform):** Size 256 for speech

Emotion Mapping Rules

 **EXCITED:** High Energy (>70) + High Pitch (>200 Hz)

 Indicates: Enthusiasm, high energy, positive excitement

Confidence: 75-90%

 **HAPPY:** Medium Energy (>50) + Medium Pitch (>150 Hz)

 Indicates: Pleasure, contentment, positive mood

Confidence: 65-85%

 **SAD:** Low Energy (<30) + Low Pitch (<120 Hz)

 Indicates: Sadness, depression, low mood

Confidence: 60-80%

 **ANXIOUS:** High Spectral Centroid (>2000 Hz)

 Indicates: Tension, worry, stress, nervousness

Confidence: 55-75%

 **NEUTRAL/CALM:** Balanced Features

 Indicates: Normal speech, relaxed, peaceful

Confidence: 55-70%



Why These Features Work

- **Energy:** Excited/happy people speak louder; sad people speak softly
- **Pitch:** Higher pitch indicates positive emotions; lower pitch indicates sad/calm emotions
- **Spectral Centroid:** Anxious/tense speech has more high-frequency content
- **Speaking Rate:** Excited people speak faster; depressed people speak slower
- **Combination:** Multiple features together give more accurate classification